

Practice set 1

Powers, factorising and algebraic fractions

Write in the spaces

Answers · official assessment schedule extracts on the last pages

QUESTION ONE

(a) Simplify each expression, leaving your answer with positive indices.

(i) $\frac{(3y)^4}{3y^{-1}}$

(ii) $\sqrt[3]{8y^{27}}$

QUESTION THREE

(a) (i) Simplify fully $\sqrt{49y^{36}}$.

QUESTION ONE

(a) Solve $\frac{2x-3}{x+4} - 3 = 0$.

(b) (i) Factorise completely $6x^3y - 15x^2\sqrt{y}$.

(ii) Simplify fully $\frac{6x^2 - x - 12}{3x^2 - 5x - 12}$.

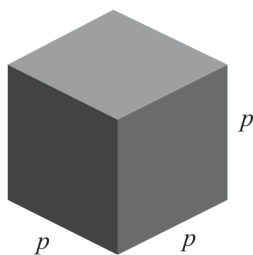
QUESTION TWO

(a) Simplify: $\frac{x^2 - x - 12}{4x + 12}$

(b) Write $\frac{5x}{x-3} - \frac{x-4}{x+2}$ as a single fraction in its simplest form.

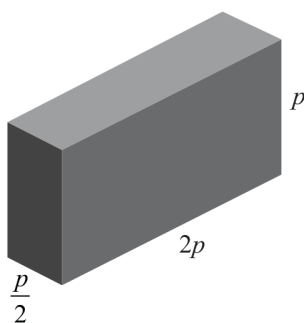
QUESTION THREE

Consider a cube with sides of p cm (where $p \neq 0$). The volume of the cube would be p^3 cm³, and the surface area of the cube would be $6p^2$ cm².



Junyang wonders if it is possible to change the dimensions of the cube to make a cuboid that still has the same volume.

- (a) First, he tries doubling the length, halving the width, and keeping the height as p , as sketched below.

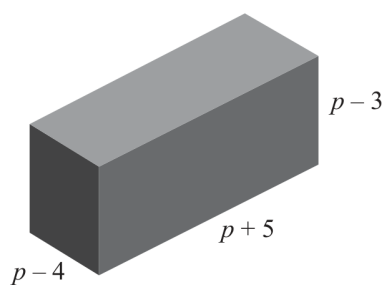


*Diagram is
NOT to scale*

Using algebra, find the volume of this cuboid.

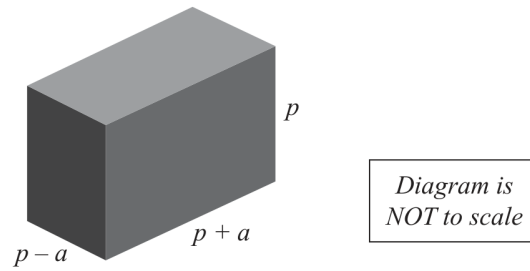
(b) The volume of the cuboid below is given by the expression: $(p - 4)(p + 5)(p - 3)$.

Expand and simplify this expression.



*Diagram is
NOT to scale*

- (c) Next, Junyang tries adding an amount, a , to the length and then taking off the same amount from the width, keeping the height the same (see below).



Are there any values of a for which the volume of the cuboid will be the same as the volume of the cube?

Official assessment schedule extract · page 1, first question

Q ONE	Expected coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)(i)	$27y^5$	Correct expression.		
(a)(ii)	$2y^9$	Correct expression.		

Official assessment schedule extract · page 1, second question

Q	Evidence	Achievement	Merit	Excellence
THREE (a)(i)	$\sqrt{49y^{36}} = 7y^{18}$	Correct response.		

Official assessment schedule extract · page 2

Q	Evidence	Achievement	Merit	Excellence
ONE (a)	$\frac{2x-3}{x+4} = 3$ $2x-3 = 3x+12$ $x = -15$	Correct solution.		
(b)(i)	$6x^3y - 15x^2\sqrt{y} = 3x^2\sqrt{y}(2x\sqrt{y} - 5)$ Accept $3x^2y\left(2x - \frac{5}{\sqrt{y}}\right)$	Obtains $3x^2(2xy - 5\sqrt{y})$	Correct expression.	
(ii)	$\frac{6x^2 - x - 12}{3x^2 - 5x - 12} = \frac{(2x-3)(3x+4)}{(3x+4)(x-3)}$ $= \frac{2x-3}{x-3}$ Don't penalise hashing a correct answer	Correct simplified fraction.		

Official assessment schedule extract · page 3

Q TWO	Expected coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$\frac{(x-4)(x+3)}{4(x+3)} = \frac{x-4}{4}$ or equivalent	Correct simplified fraction (not bashed).		
(b)	$\frac{5x(x+2) - (x-3)(x-4)}{(x-3)(x+2)}$ $= \frac{4x^2 + 17x - 12}{(x-3)(x+2)}$ or equivalent	Line 1 set up, or equivalent.	Obtains correct fraction.	

Official assessment schedule extract · pages 4 to 6

Q THREE	Expected coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	Volume of cuboid = p^3 in simplest form.	Correct answer.		
(b)	$(p - 4)(p + 5)(p - 3)$ $= (p - 4)(p^2 + 2p - 15)$ $= p^3 - 2p^2 - 23p + 60$	Correct simplified expression.		
(c)	Volume of cuboid $= p(p + a)(p - a)$ $= p^3 - pa^2$ If this is the same as the volume of the cube, $p^3 - pa^2 = p^3$ so $pa^2 = 0$ Thus $a = 0$ is the only solution (since $p \neq 0$). OR that there are no [non-trivial] solutions. Hence the cuboid never has the same volume as the cube unless you leave the sides unchanged. Or equivalent.	Correct working to the point of equating the two volumes.	Makes correct conclusion with valid working and reasoning.	